

PANORAMA IMAGE SYSTEM

Multi-Sequence 360° Mini Product Requirements Document

A production-ready method for coherent warehouse panoramas, hazard staging, hotspot readiness, and visual QA

Product	Safe360
Document	PAN-PRD-001
Version	1.0 - 27 August 2026
Status	Implementation-ready; safety validation pending

How the sequence works

ONE WORLD BIBLE → SIX LINKED SCENES → IMAGE LOCK → HOTSPOT COORDINATES → VIEWER QA

Core decision: generate continuity first and hazards second. Coordinate work begins only after the visual asset is locked.

1. Product definition

Purpose. Define a repeatable way to create a visually consistent sequence of 360-degree warehouse panoramas for interactive hazard-perception training. This specification covers the image system only: scene planning, generation, continuity, hotspot readiness, optimization, and visual acceptance.

Product outcome. A learner should feel they are moving through one believable warehouse - orientation area, loading bay, storage aisles, vehicle zone, packing/dispatch, and emergency/fire area - even though each panorama is a separate image.

Primary users. Content designer, panorama artist or AI operator, safety reviewer, hotspot author, QA tester, and the learner consuming the finished sequence.

Safety boundary. Images are training prototypes until a qualified workplace-safety reviewer confirms that each depicted hazard is accurate, visible, unambiguous, and paired with correct site-specific guidance.

2. Scope and success measures

In scope. Six related equirectangular panoramas; a shared visual world bible; controlled camera and lighting; scene-specific hazard staging; prompt templates; seam and projection QA; hotspot coordinate planning; local web delivery; revision and approval records.

Out of scope. Video 360, stereoscopic VR, real-time 3D navigation, photogrammetry, multiplayer, automatic safety certification, and UI overlays baked into source images.

Success measures. 100% of images open in the viewer; every source image is 2:1; no blocking seam or projection defect; all required hazards are visible at intended zoom; the sequence is recognizably one facility; every approved hotspot maps to a real visual target; median compressed load meets the agreed device budget.

3. The multi-sequence model

Sequence architecture. Treat the set as linked rooms in one world, not six unrelated illustrations. Each image has a scene ID, narrative purpose, entry direction, exit direction, repeated visual anchors, required hazards, and an approval state.

Continuity anchors. Repeat the same architecture language, floor markings, rack colors, PPE palette, lighting temperature, wall materials, signage style, pallet types, and equipment family. Reuse two or three memorable anchors - for example a teal safety rail, yellow floor lanes, and gray-blue racks - across adjacent scenes.

Camera contract. Use a central eye-level camera at approximately 1.6 m, level horizon, identical projection, stable verticals, and a consistent initial viewing direction. Avoid placing critical hazards directly at the top/bottom poles or across the left/right seam.

Transition contract. The exit region of scene N should visually suggest the entry region of scene N+1. Match door type, corridor width, light direction, floor markings, and nearby equipment. The images do not need pixel-perfect portals, but their spatial story must agree.

4. Functional requirements

PAN-FR-001. The system shall define one master world bible before producing individual scenes.

PAN-FR-002. Every final panorama shall use an equirectangular 2:1 projection and a unique stable filename.

PAN-FR-003. Every scene brief shall identify purpose, camera orientation, continuity anchors, required hazards, exclusions, and transition cues.

PAN-FR-004. The sequence shall use consistent facility identity, materials, lighting, equipment, and color language.

PAN-FR-005. Hazards shall be visible, plausible, separated enough for individual hotspots, and never dependent on unreadable generated text.

PAN-FR-006. No hotspot marker, label, score, instruction, logo, or UI element shall be baked into a panorama.

PAN-FR-007. Each image shall be reviewed in a real 360 viewer for seam, poles, horizon, distortion, and hotspot visibility.

PAN-FR-008. Hotspot yaw and pitch shall be captured only after the final image version is locked.

PAN-FR-009. Every revision shall retain prompt, generator/model, date, source version, reviewer, and change reason.

PAN-FR-010. Approved images shall be optimized and served locally with a friendly loading/error state.

5. Scene sequence specification

Scene	Visual and learning brief
S00 - Orientation/Tutorial	Calm induction corner. One obvious practice condition. Establish the facility palette, camera behavior, and learner confidence. Exit cue points toward the loading operation.
S01 - Loading Bay	Dock doors, trailer interface, pallet staging, and vehicle movement. Emphasize edge protection, unstable loads, spills, visibility, and pedestrian exposure. Transition toward internal storage.
S02 - Storage Aisles	Tall racks and long sightlines using the same rack/floor system. Emphasize damaged racking, poor stacking, obstruction, falling-object risk, and housekeeping. Transition toward the shared traffic zone.
S03 - Forklift/Pedestrian Zone	Open intersection linking aisles to dispatch. Emphasize segregation, blind corners, speed, visibility, parked equipment, and pedestrian behavior. Transition toward packing benches.
S04 - Packing and Dispatch	Workstations, conveyors, cartons, and outbound staging. Emphasize ergonomics, guarding, sharp edges, clutter, cables, blocked routes, and load visibility. Transition toward the emergency route.
S05 - Emergency and Fire	Exit route, fire equipment, electrical points, and waste control in the same facility language. Emphasize blocked access, combustible waste, obscured signage, narrowed egress, and damaged equipment.

6. Production workflow

Step 1 - Build the world bible. Lock facility type, dimensions, palette, lighting, camera height, lens feel, repeating assets, cleanliness baseline, PPE, prohibited content, output size, and naming convention.

Step 2 - Make a scene manifest. For each scene list required visible hazards, continuity anchors, transition cues, camera heading, safe negative space for controls, and elements that must not appear.

Step 3 - Generate a clean base. Create the environment and continuity first. Do not overload the first generation with every hazard. Approve geometry, identity, seam potential, and camera before adding complex conditions.

Step 4 - Add hazards in controlled passes. Add hazards by zone or category while preserving the locked base. Prefer image editing/inpainting over complete regeneration when correcting one local condition.

Step 5 - Projection QA. Inspect as a flat 2:1 image and inside Marzipano. Rotate across the seam; inspect poles; confirm a level horizon, coherent scale, and no duplicated or melted equipment.

Step 6 - Safety and learning QA. Confirm each intended target is visible, recognizable without a label, plausible, not accidentally contradicted, and suitable for one unambiguous question.

Step 7 - Lock and coordinate. Freeze the image checksum/version, capture yaw/pitch, test hotspots at desktop and tablet widths, then export an asset manifest.

Step 8 - Optimize and publish. Retain a lossless master. Export a web derivative, verify quality and load behavior, deploy locally, and keep a rollback copy.

7. Prompt system

Master prompt. Create a seamless full 360-degree equirectangular panorama of one modern warehouse facility, exact 2:1 aspect ratio, photorealistic professional safety-training style, central eye-level camera at 1.6 m, level horizon, coherent wraparound geometry, neutral daylight-balanced industrial lighting, gray-blue racking, yellow floor lanes, teal safety rails, clean realistic materials, no text overlays, no hotspot markers, no logos, no fisheye circle, no split panels, no duplicated people or equipment, no gore.

Scene addendum template. SCENE [ID/TITLE]. Narrative purpose: [purpose]. Preserve world-bible architecture, palette, camera, and lighting. Required visible zones: [zones]. Required hazards: [numbered hazards with location and visual evidence]. Continuity from previous scene: [anchors]. Transition toward next scene: [cue]. Reserve readable negative space near [HUD region].

Negative prompt. Avoid cubemap faces, cylindrical projection, cropped 180-degree view, warped horizon, broken walls, mirrored signage, unreadable safety text, extra limbs, fused pallets, duplicated forklifts, impossible rack geometry, hazards hidden behind objects, critical targets on the seam/poles, and baked UI.

Repair prompt. Edit only [bounded region]. Preserve every other pixel-level scene relationship, camera position, projection, lighting, palette, and existing approved hazards. Correct [specific defect]. Do not add new people, text, equipment, or hazards.

8. Technical image specification

Projection. Equirectangular, full 360 x 180 degrees, exact width:height ratio of 2:1.

Master. PNG or high-quality TIFF, sRGB, recommended 8192 x 4096 where the generation/capture pipeline supports it. Preserve this file as the immutable source master.

Web derivative. JPEG or WebP at an empirically tested size, commonly 4096 x 2048 for a lightweight prototype. Choose quality from visual tests, not a fixed number. Keep a lossless version for future re-export.

Naming. pan-[sequence]-[scene-slug]-v[major.minor]-[state].[ext], for example pan-03-forklift-zone-v1.2-approved.webp.

Hotspot coordinates. Yaw range $-\pi$ to $+\pi$; pitch range approximately $-\pi/2$ to $+\pi/2$. Store coordinates separately from the image and include image version/checksum.

Delivery. Use local same-origin assets, width/height metadata, loading state, fallback message, next-scene prefetch only, and cache-busting versioned filenames.

9. Acceptance criteria

PAN-AC-01. Six planned scenes exist and tell one coherent warehouse journey.

PAN-AC-02. Every approved file is exactly 2:1 and renders through 360 x 180 degrees without a missing region.

PAN-AC-03. No obvious left/right seam, broken horizon, unacceptable pole distortion, duplicated major object, or unreadable critical target is present.

PAN-AC-04. All planned hazards are visible and individually selectable at the supported desktop and tablet viewports.

PAN-AC-05. No critical hotspot overlaps another hotspot, viewer control, or modal trigger at the initial and expected viewing positions.

PAN-AC-06. A safety reviewer signs off the depiction and a learning reviewer signs off target ambiguity.

PAN-AC-07. Prompts, generation metadata, source master, web derivative, checksum, coordinates, and review decision are recorded.

PAN-AC-08. Missing or slow assets produce a clear recoverable state rather than a blank viewer.

10. Risks and controls

Continuity drift. Control with a locked world bible, reference images, shared seed/style where supported, and image editing instead of full regeneration.

Projection defects. Control with explicit equirectangular language, 2:1 checks, seam rotation, pole inspection, and rejection before hotspot work.

Hazard ambiguity. Control with one visual cause per learning target, minimum separation, a hazard evidence note, and reviewer sign-off.

Generated text/errors. Do not depend on AI-rendered words. Add required readable signage through a controlled post-production process and recheck projection.

Coordinates become stale. Bind coordinates to asset version/checksum and recapture after any crop, resize that changes mapping, or visual regeneration.

Large downloads. Use measured derivatives, caching, lazy loading, and next-scene prefetch while retaining the master offline.

11. Definition of done

Release checklist. World bible approved; six scene briefs complete; masters and web derivatives present; aspect ratio and file integrity pass; seam/pole/horizon checks pass; hazards and transitions reviewed; safety status recorded; image versions locked; hotspot yaw/pitch captured and device-tested; provenance and checksums stored; loader/fallback tested; rollback copy available.

Appendix A - Scene brief template

Scene ID and title: _____

Narrative purpose: _____

Previous-scene continuity anchors: _____

Next-scene transition cue: _____

Camera heading and initial view: _____

Required visible zones: _____

Required hazards and visual evidence: _____

Exclusions / negative constraints: _____

Safe negative-space regions: _____

Master and derivative filenames: _____

Image checksum/version: _____

Safety reviewer / date / decision: _____

Hotspot reviewer / date / decision: _____

Approval note: do not publish operational training solely from generated imagery. Validate against applicable law, site procedures, equipment guidance, and qualified safety expertise.